Project Design Phase
Solution Architecture

Date	19 July 2026
Team ID	
Project Name	Nightjar & Co. – MERN BookStore
Maximum Marks	4 Marks

Solution Architecture

Overview

Nightjar & Co. – MERN BookStore follows a three-tier architecture consisting of the presentation layer, application layer, and database layer. The React frontend provides a responsive user interface where users can browse books, search by category, manage their cart, and authenticate securely. Requests are sent to the Express.js and Node.js backend through REST APIs. The backend processes business logic, validates requests, manages JWT authentication, and interacts with the MongoDB database to store and retrieve application data.

This architecture provides clear separation of concerns, making the application scalable, maintainable, and easy to extend with additional features.

Data Flow

1. The user accesses the React frontend through a web browser.
2. The frontend sends API requests to the Express.js backend.
3. The backend validates requests and authenticates users using JWT.
4. Business logic is executed for operations such as login, searching books, cart management, and order handling.
5. MongoDB stores and retrieves user, book, and order data.
6. The backend returns JSON responses to the frontend, where the user interface is updated.

Technologies Used

Layer	Technology
Presentation Layer	React.js (Vite), HTML, CSS, JavaScript
Application Layer	Node.js, Express.js
Authentication	JWT, bcrypt
Database Layer	MongoDB, Mongoose
API Communication	REST API (JSON)

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